GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

Scale Factor: Perimeter and Area

Question Bank - Additional 5.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the copy's perimeter or area.

1. Perimeter 26 cm, $k = 3$.

78 cm

2. Area $9\,\text{cm}^2$, $k = 3$.

81 cm²

9 × 9.

3. Area $100\,\text{cm}^2$, $k = \frac{1}{2}$.

25 cm²

100 × ¼.

4. A 4 cm by 5 cm rectangle, $k = 1.5$. Perimeter and area of the copy.

P = 27 cm, A = 45 cm²

18 × 1.5; 20 × 2.25.

For questions 5 through 7, find k , or the ratio asked.

5. Areas $11\,\text{cm}^2$ and $99\,\text{cm}^2$. Find k .

k = 3

99 ÷ 11 = 9 = k^2.

6. Areas $32\,\text{cm}^2$ and $2\,\text{cm}^2$, original to copy. Find k .

k = ¼

2 ÷ 32 = 1⁄16 = k^2, so k = ¼.

7. Perimeters 10 cm and 50 cm, original to copy. What is the area ratio?

25

k = 5; k^2 = 25.

Choose one.

8. A figure with area $4\,\text{cm}^2$ is scaled by $k = 6$. What is the copy's area?

- ○ A. $48\,\text{cm}^2$
- ● **B.** $144\,\text{cm}^2$
- ○ C. $24\,\text{cm}^2$
- ○ D. $10\,\text{cm}^2$

B

$4 \times 6^2 = 4 \times 36$.